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Code No: 135SC

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
OPERATING SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

User Papers March-2024

PART – A

(25 Marks)

1.a)	Define an operating system.	[2]
b)	What are the objectives of OS?	[3]
c)	Distinguish between a process and a thread.	[2]
d)	What is critical section?	[3]
e)	Define thrashing.	[2]
f)	What is meant by paging and segmentation?	[3]
g)	Write about file sharing.	[2]
h)	What is the purpose of unlink()?	[3]
i)	What is mutual exclusion?	[2]
j)	What is wait-for-graph?	[3]

PART – B

(50 Marks)

2.a) Explain various operating system services.
b) Draw and explain the layered architecture of an operating system. [5+5]

OR

3.a) Write about operating system role in storage management and security.
b) What is a system program? Give some examples. [6+4]

4. Perform preemptive and non-preemptive SJF scheduling on following data and compute average waiting time and average turnaround time. [10]

Process ID	Arrival time	Burst time
P01	0	4
P02	1	5
P03	3	7
P04	5	1
P05	6	9

OR

5.a) Distinguish between unnamed pipes and named pipes.
b) List and explain various scheduling criteria. [5+5]

6. Explain various memory allocation techniques with clear illustrations. [10]

OR

7. Simulate FIFO, LRU and optimal page replacement algorithms on following reference string and compute the number of page faults for a frame size of 3 and assuming the frame is initially empty. [10]

1,2,3,1,5,6,7,2,3,4,2,4,6,7,2,3,4,1

8. List and explain various file directory structures. [10]

OR

9.a) Distinguish between SCAN and C-SCAN disk scheduling algorithms.
b) Explain about file system mounting. [5+5]

10.a) What are the principles of protection?
b) What are the various deadlock prevention strategies? Explain them. [4+6]

OR

11.a) What is protection? What are the goals of protection?
b) What are various methods for recovering from deadlock? Explain. [4+6]

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Used Papers March-2024